

few human activities are responsible for this influence on the global nitrogen cycle: combustion of fossil fuels and biomass, and production and use of nitrogenous fertilizers. These activities have contributed to major changes to the global nitrogen cycle, and in some cases significant environmental problems.

Glossary

Denitrification Biological process by which aqueous NO_3^- is converted into mainly N_2 and some N_2O .

Dry deposition Removal of a chemical species from the atmosphere by direct interaction with the ground.

Industrial nitrogen fixation Conversion of N_2 to a nitrogen compound, either intentionally via the Haber process generating NH_3 , or unintentionally as during fossil fuel combustion to generate NO .

Lifetime Time constant ($1/e$) associated with a specific chemical or removal process. For atmospheric compounds, the lifetime refers to the time when 37% ($1/e$) of the original amount remains.

Nitrogen fixation Any process that converts N_2 into a nitrogen compound. In biological nitrogen fixation N_2 is converted into NH_3 or organic compounds via an enzyme-catalyzed process.

Photochemical smog Mix of photochemically generated gases and aerosols often found in warm, sunny urban environments. Key in this process is the accumulation of O_3 , which is associated with health impacts and vegetation damage.

Wet deposition Removal of a chemical species from the atmosphere via rain or fog.

See also

Carbon Dioxide. Global Change: Ozone Trends. Soot. Stratospheric Chemistry and Composition: Reactive Nitrogen (NO_x and NO_y).

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Sulfur Cycle

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Introduction

The sulfur cycle transfers enormous amounts of this biologically important element through the atmosphere every year. Despite the size of natural reservoirs,

human activities have had a remarkable effect on the sulfur cycle, perhaps most notably in the production of acid rain. The Earth's surface is a source of windblown dusts, but this is influenced by grazing activities and desertification. Large amounts of sulfur are mobilized by mineral extraction and fuel use. Fossil fuel refining and combustion are the major human emissions to the atmosphere, which exceed the natural sources. The reduction in anthropogenic emissions from North

America and Europe is countered by growing emissions from Asia and may represent a special challenge for the twenty-first century.

Sulfur

The sulfur cycle is one of the major elemental cycles. Sulfur has an enormous span of oxidation states: from sulfides in the $-II$ state through to sulfates in the $+VI$ state. Both sulfides and sulfates are frequently found as minerals, which include the innumerable sulfides of metals or sulfates such as that of calcium (gypsum) and/or barium (barite). Sulfur is also found in its elemental state. Rhombic and monoclinic crystalline forms both contain S_8 rings. Liquid sulfur undergoes sharp changes in properties (color, viscosity, specific heat) as the temperature rises above the melting point (112.8°C for rhombic sulfur), with complex changes in the ring structure. Sulfur is essential to living organisms because it is incorporated into amino acids, proteins, and other biomolecules.

There is a naturally occurring radioisotope, sulfur-35, produced by the interaction of cosmic rays with argon in the upper atmosphere. It has a radiochemical half-life of 87 days and is rapidly oxidized to sulfur dioxide (SO_2) and sulfate, which is ultimately transferred to the lower atmosphere and removed in rain.

Evolution of the Sulfur Cycle

Sulfur is a common element with a high cosmic abundance half that of silicon (a frequent marker element). Concentrations of sulfur are much reduced in the Earth's crust because of downward separation when it is segregated in heavy sulfide magmas. Places where sulfur is plentiful are usually the result of intrusions that occurred in the earliest stages of the Earth's history. The element remains very much associated with volcanic activity. The Earth formed through gradual accretion of infalling material during Hadean time (4.5 to 3.8 billion years ago). Accretion left the mantle extensively molten. Gradually the Earth's surface cooled and solid rock formed, probably prior to 3.8 billion years ago. Metallic iron influenced the oxidation state of outgassed volatiles and hydrogen sulfide would have made an important contribution to the early sulfur budget. This differs from the current dominance of volcanic gases by sulfur dioxide. On the Archean Earth (3.8 to 2.5 billion years ago) hydrogen sulfide could be removed from the atmosphere through dissolution and precipitation as iron sulfide from the oceans or destruction by photochemical processes or lightning in the atmosphere.

Within the early oceans the concentration of sulfate would have been controlled by the provision of oxidizing materials from the atmosphere. Sulfate could not have risen to high concentrations, while the hydrogen content of the atmosphere was maintained in excess of 0.0001 atm through rapid volcanic input.

Atmospheric oxygen remained considerably less abundant through the Archean (3.4 to 2.8 billion years ago) than today. Nevertheless at the beginning of the Archean, anaerobic photosynthesis provided a weak source of sulfate to the global ocean. Photodissociation of water provided a limited amount of oxygen that was supplemented with oxygen from oxygenic photosynthesis about 2.8 billion years ago. This would probably lead to atmospheric oxygen concentrations high enough to oxidize the sulfur in pyrite to sulfate. Bacterial reduction of sulfate to sulfides in marine sediments is a good indicator of increasing sulfate in sea water, with early indications found between 2.7 and 2.5 billion years ago. The sedimentary sulfur isotope record suggests that there were low concentrations of seawater sulfate and oxygen in the early Archean and atmosphere, respectively. Sulfate concentrations in sea water were probably less than a millimolar until the substantial oxygen increases of the early Proterozoic (2.5 to 0.54 billion years ago).

Volcanic Sources of Sulfur

Although elemental sulfur occurs in volcanic gas, perhaps as the dimer, most is found as either SO_2 or hydrogen sulfide (H_2S). The equilibrium calculations of the composition of volcanic gases show oxidized systems that contain SO_2 , and smaller amounts of S_2 and H_2S . More reduced systems may have COS and H_2S in greater relative abundance.

Some fluxes of sulfur gases from volcanoes occur as fairly continuous small releases from fumaroles into the atmospheric boundary layer. Violent volcanic eruptions represent a more sporadic but potentially much larger source of sulfur. The Tambora eruption of 1815 released some 50 Tg of sulfur, but this was exceptionally large (by comparison the Pinatubo eruption of 1991 released perhaps only one-fifth as much). Such explosive volcanism can inject materials into the atmosphere with such force that the sulfur compounds can be driven into the stratosphere (Figures 1 and 2). These gases are slowly converted to sulfate aerosols with a residence time of about a year in the stratosphere. Larger ash particles sediment out more quickly. Sulfur condenses as sulfur crystals on rocks near volcanic vents. This has provided a source of sulfur since ancient times and the risky activity of mining volcanoes for sulfur continues to the present day.

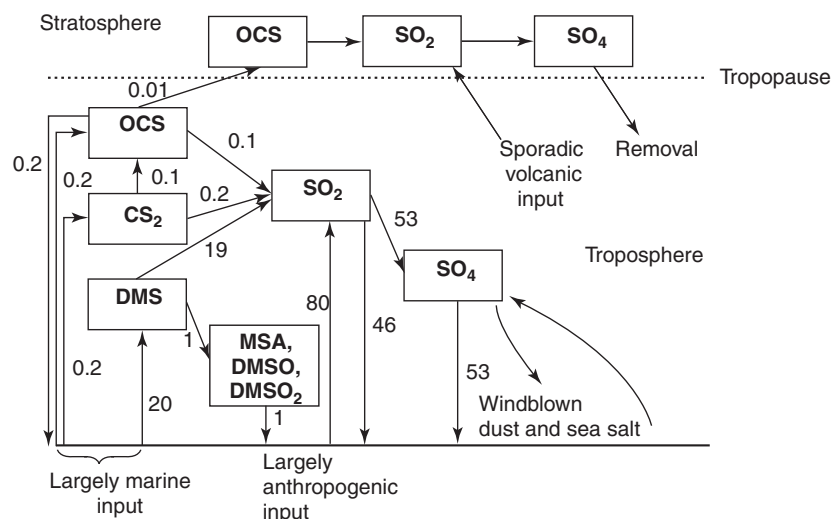


Figure 1 Atmospheric sulfur cycle with fluxes in Tg(S) year^{-1} . The reduced sulfur compounds DMS, OCS, and CS_2 come largely from marine processes. In years with much explosive volcanism there can be large sources of SO_2 directly to the stratosphere. (Adapted with permission from Rodhe (1999).)

The variability and the shifting balance between fumarolic and explosive injections make globally averaged emission estimates difficult. Over the last quarter of the twentieth century the annual volcanic emissions have varied by a factor of three, but on average they were $7 \text{ Tg(S) year}^{-1}$ as SO_2 and about $2.6 \text{ Tg(S) year}^{-1}$ as H_2S . The volcanic emissions of carbon disulfide and carbonyl sulfide are an order of magnitude lower than H_2S .

Recently it is clear that the sulfuric acid droplets in the stratosphere (Figure 2) can enhance ozone

depletion by providing sites for heterogeneous reactions. This can lead to a reduction in the thickness of the ozone layer after volcanic eruptions.

Marine Sulfur Sources

Present-day sea water has very high concentrations of sulfur, largely because of its high solubility as the sulfate. Sulfate is found at 0.028 mol l^{-1} and is thus the most abundant anion apart from chloride. Hydrothermal processes at midocean ridges are able to reduce sulfates to sulfides. As sea water evaporates it first precipitates calcium and magnesium carbonates, and calcium sulfate, gypsum. When sea water is trapped within basins in arid climates, it can lead to extensive deposits of evaporite minerals. In the geological past these shallow basins were more extensive than at present and gave rise to very extensive evaporite beds.

Sea spray can be blown directly from the sea surface during high winds. The particles of salt from this process tend to be large and short-lived. Smaller particles are produced during bubble bursting. The very finest arise when the cap of a surfacing bubble shatters, giving film drops that dry into salt crystals of 5–50 μg . Along with these are larger jet drops that emerge as the bubble cavity collapses and give particles of about 0.15 μg . Such processes lead to the production of around a gigagram of sea salt particles each year, of which 30 Tg would be sulfur in the form of sulfate. Thus high chloride and sulfate concentrations are found in coastal rainwater.

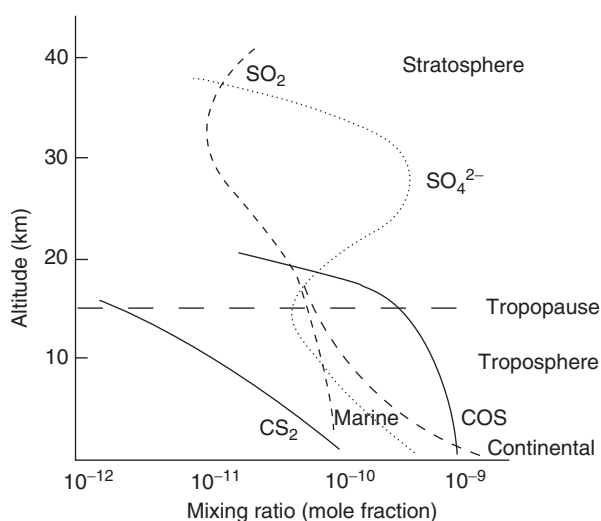
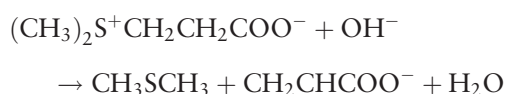


Figure 2 Vertical profile of sulfur species. The sulfate aerosol layer in the stratosphere has important effects on temperatures and ozone depletion. The marine and continental profiles for sulfur dioxide are different in the troposphere.

H₂S is extensively produced from sulfate reduction in tidal flats and marshes. This is not surprising given the high concentration of sulfate in sea water. H₂S had long been associated with biological activities and was always assumed to be an important source of reduced sulfur to the atmosphere. Anaerobic environments are protected from invasion of oxygen through burial or lying underwater. This limited exchange of oxygen can also restrict the loss of biogenic H₂S to the atmosphere. Concentrations are typically in the range 30–100 ppt in remote air, although on tidal mud flats or in volcanic regions they can be much higher. H₂S oxidizes rapidly in sea water (c. 10 min), so any of this gas found in marine air would have to be produced close to the sea surface.

A preoccupation with H₂S has meant that early studies overlooked other reduced sulfur gases. The most notable was dimethylsulfide (DMS or CH₃SCH₃) identified first by James Lovelock in the 1970s. DMS is the largest natural source of reduced sulfur gases. It is produced through the activities of phytoplankton and most arises from dimethylsulfonium propionate ([(CH₃)₂S⁺CH₂CH₂COO[−]] or DMSP) which leaks from aging or ruptured cells. The bacterial decomposition of DMSP in sea water cleaves it into DMS and acrylic acid:



High concentrations of DMS are found in areas where microorganisms are abundant, but the amounts released vary with species and the stage of their life cycle. Seasonal maps of DMS in sea water in the oceans have become quite detailed, reflecting its biological origin. DMS is released from sea water as it is relatively insoluble and readily transfers to the atmosphere. The fluxes to the atmosphere are dependent not only on concentration but also on wind speed. However, there remains an uncertainty of a factor of two with the estimated emissions of 20 Tg(S) year^{−1}. Some DMS in sea water is oxidized, probably to dimethylsulfoxide (DMSO, CH₃SOCH₃). This may be a photosensitized or biological oxidation, while some DMSO is produced directly from biological activities.

DMSO is more soluble than DMS, and so is unlikely to degas from sea water in large amounts. Other alkyl sulfides are more likely to be released from the ocean, but their abundances and thus fluxes are much smaller. Methanthiol (CH₃SH) and dimethyl-disulfide (CH₃SSCH₃) are perhaps the best known of these.

The oceans are also a source of carbon disulfide (CS₂) which is released from the ocean at about

0.2 Tg year^{−1}. CS₂ is produced by photosensitized reactions in sea water. Laboratory irradiations have confirmed that the amino acids cysteine and cystine are efficient precursors of CS₂. Biological production of CS₂ has also been observed in some marine phytoplankton.

CS₂ can be easily oxidized and contributes to the production of carbonyl sulfide (OCS). There are also photochemical mechanisms for the production of OCS from organic matter in the oceans. The emissions show a diurnal cycle due to the photochemical origin, but the fluxes from the ocean are much smaller than DMS. Nevertheless OCS is the most abundant reduced sulfur gas in the atmosphere as it is very stable in the troposphere (25 year residence time) and thus able to build up to substantial concentrations (0.5 ppb).

The low-temperature oxidation products of atmospheric DMS, most notably DMSO, dimethylsulfone (CH₃SO₂CH₃), and methanesulfonic acid (MSA, CH₃SO₃H), have been found in the Antarctic atmosphere typically at 1.5, 1.3, and 30 ppt, respectively. They may all be found in rainwater, but MSA is especially soluble and so is largely associated with aerosol particles. At times MSA is almost as abundant as non sea salt sulfate (NSS), but under warmer conditions it may only be a 50th that of NSS, the major oxidation product.

These marine sources of reduced sulfur gases can be important as a source of sulfur to the continents. The gypsum accumulations of the hyper-arid Central Namib Desert derive from non sea salt sulfur, in particular oxidation products of marine DMS.

Terrestrial Sulfur Sources

Windblown dusts are important as a source of sulfate in the atmosphere. The largest of these often occur from evaporitic deposits. Dusts from the margins of the Aral Sea (0.1–1 Tg year^{−1}) and arid states of the southern United States are notable source regions.

Biogenic sulfur emissions from land have also been recognized as making an important contribution to the global budget, but the fluxes have been more difficult to establish. Many of the gases that originate from biological activity in the oceans are also emitted from the continents, although the quantities are far less certain. Growing crops and biochemical activities in wetlands, especially in salty coastal environments, give predominantly DMS, H₂S, and methyl sulfide, with generally much lower amounts of OCS and CS₂. Tundra, while seemingly a likely source, has been found to be a low emitter of reduced sulfur because of low sulfate amounts in the water.

Biological Sinks for Sulfur

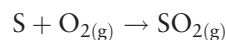
Vegetation can remove sulfur compounds from the atmosphere. Carbonyl sulfide removal by plants is the best known and probably represents a dominant sink for tropospheric OCS. The fate of some oxidation products of DMS is not clear. Some biochemical studies suggest that they may degrade biologically. Facultatively methylotrophic species of *Hyphomicrobium* and *Arthrobacter* seem to be able to produce enzymes necessary for a reductive/oxidative pathway for DMSO and dimethylsulfone (although the former appears to be more generally utilizable). Methanesulfonic acid is stable to photochemical decomposition in the atmosphere and its fate on land has been puzzling. A range of terrestrial methylotrophic bacteria that appear common in soils may mineralize MSA to carbon dioxide and sulfate.

Anthropogenic Sulfur Sources

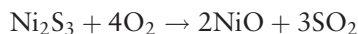
The most pernicious source of human sulfur emissions has been from the combustion of high-sulfur coals. This began in thirteenth-century London with the depletion of nearby wood supplies and increased through the centuries that followed until it reached its peak within the early twentieth century in Europe and North America. In Asia, particularly, emissions have continued to grow with the enormous pressure of industrial development. In the North Atlantic sector there has been a general decline in sulfur emissions with a shift away from coal as a fuel, in all but extremely large industrial plants, as a response to the problems of urban air pollution and acid deposition. It is estimated that in south and eastern Asia more than 10^7 km^2 of land will be subjected to sulfur deposition greater than $1 \text{ g(S) m}^{-2} \text{ year}^{-1}$ by AD 2020.

Sulfur in coal exists half as pyrites, which is relatively easy to remove, but the rest is organically bound. This makes it hard to remove except by methods such as coal gasification or in stacks after emission. Sulfur can be removed by catalytic hydrodesulfurization from residual oil, but it leads to fuels that tend to become waxy at low temperature.

The combustion of fuels leads to the release of SO_2 in a simple but effective oxidation:



Many refining and extractive processes release large amounts of air pollution containing SO_2 . For example, sulfide ores have been roasted in the past with uncontrolled emissions:



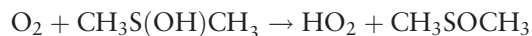
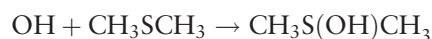
The SO_2 released often destroyed large tracts of vegetation downwind from smelters.

Various techniques are used to remove sulfur from stack gases. The use of lime (calcium hydroxide) or limestone (calcium carbonate) slurries to absorb SO_2 is widely adopted. The main product, calcium sulfate, is not seen as an environmentally hazardous by-product, although it can be contaminated with trace metals. However, the amounts of lime required can be extremely large. There are often problems where these are mined from attractive sites of great ecological and recreational value. Regenerative desulfurization processes such as the Wellman-Lord procedure absorb SO_2 into sodium sulfite solutions converting them to sodium bisulfite. The SO_2 is later degassed and can be used as a feedstock for the chemical industry (i.e., the production of sulfuric acid).

Catalytic converters in vehicles have been used to remove nitrogen oxides, carbon monoxide, and hydrocarbons from exhaust streams. However, under fuel-rich driving cycles (i.e., lots of accelerating and decelerating), hydrogen gas is produced in the exhaust. Three-way catalysts contain cerium dioxide, which stores sulfur from gasoline under driving conditions as cerium sulfate. Reduction of the cerium sulfate by hydrogen gas allows the formation of H_2S . This can create a noticeable odor, where traffic is heavy.

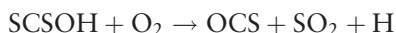
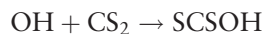
Atmospheric Sulfur Chemistry

Typically the gases we have been discussing are in a somewhat reduced form in the atmosphere and undergo oxidation. DMS is oxidized in an oxygen-rich atmosphere via OH attack:



DMSO can yield dimethylsulfone. Methanesulfonic acid is an important product of DMS oxidation although the process is not entirely clear. It has been suggested that this arises from CH_3SO_2 produced in a sequence of reactions that follow hydrogen abstraction by OH from DMS. Once fragmented, the S from DMS can oxidize to SO_2 and sulfuric acid. Lower temperatures as found at high latitude appear to favor the production of larger proportions of DMSO and MSA rather than the more complete oxidation through to SO_2 . The oxidation of DMS typically takes place at time scales of about a day. These processes are an important source of sulfuric acid over the ocean and contribute to the production of cloud condensation nuclei. Sulfur in this form is usually called non sea salt sulfate to distinguish it from the sulfate from sea salt.

CS₂ reacts in the atmosphere with a 7 day residence time.

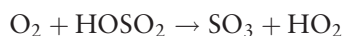
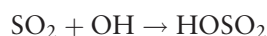


OCS oxidizes only slowly:



However, in the stratosphere the molecule can photofragment into CO and S which can ultimately oxidize to contribute to sulfate particles.

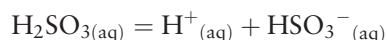
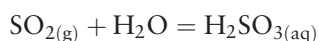
Homogeneous oxidation of SO₂ takes place through OH attack:



The sulfur trioxide reacts very rapidly with water to form sulfuric acid. This can be a significant pathway, but if liquid water is present heterogeneous mechanisms yield most of the sulfate.

Chemistry of Sulfur in Rain and Cloud Water

The interaction of SO₂ with water is responsible for much of its removal from the atmosphere. SO₂ is not especially soluble in water, but subsequent equilibria increase the partition into cloud water at typical pH values.



$$K_H = m\text{H}_2\text{SO}_3/p\text{SO}_2, \quad 5.4 \text{ mol l}^{-1} \text{ atm}^{-1} \text{ at } 15^\circ\text{C}$$

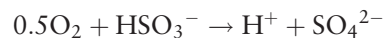
$$K' = m\text{H}^+ m\text{HSO}_3^- / m\text{H}_2\text{SO}_3,$$

$$0.027 \text{ mol l}^{-1} \text{ at } 15^\circ\text{C}$$

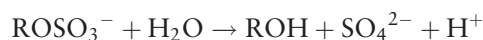
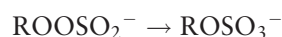
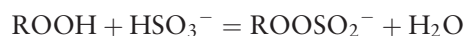
where *m* and *p* refer to the molarity and pressure of aqueous and gaseous species, respectively.

SO₂ dissolves in water as (SO₂)_{aq} 'sulfurous acid' (H₂SO₃). It is usually sufficient to treat only the first dissociation to the bisulfite anion (HSO₃⁻) as the subsequent dissociation to the sulfite ion is usually small in cloud droplets. The concentrations expressed here in molar or molal units should be as activities, but in rainwater the deviations from ideality are small. At pH 5.4 in a typical cloud with a gram of liquid water in each cubic meter, SO₂ will partition equally into both phases. Once dissolved, sulfur dioxide can be oxidized, but oxidation by dissolved oxygen is typically

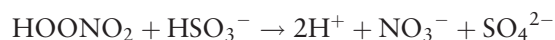
rather slow. The production of sulfuric acid, which is much stronger, leads to acidification.



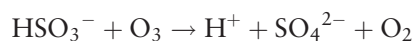
This simple oxidation can be catalyzed by iron, manganese, and other transition metals in the atmosphere, but in unpolluted rainfall the metal concentrations are likely to be low. Here more typically the reaction proceeds with oxidants such as dissolved hydrogen peroxide (or other atmospheric peroxides) and ozone. The hydrogen peroxide route is a particularly significant one as the reaction is faster in acid solution. This means that it does not slow as the system becomes more acidic with the production of sulfuric acid. This oxidation can be represented as follows:



Hydrogen peroxide is an especially important droplet phase oxidant because the gas is very soluble in water and readily produced in atmospheric solutions by photochemical mechanisms. Other oxidants produced in solution, such as peroxyntic acid, appear to be important:



The ozone-driven oxidation is written as



and although ozone is abundant in the atmosphere it is not very soluble.

In general, the overall oxidation in the remote atmosphere takes 2–4 days, but under some conditions it can be much faster. In urban areas plentiful supplies of transition metal catalysts and oxidants can enhance the rates considerably, especially if there are alkaline materials from fly ash or ammonia to neutralize the growing acidity of droplet phases which limits SO₂ solubility.

Land surfaces, particularly those covered by vegetation, represent a further removal process as SO₂ is dry deposited to these surfaces. Although we use the term dry deposition, the vegetated surfaces act as if they are wet and gas exchange takes place with effectively wet surfaces on or inside leaves. Such dry deposition is most significant where SO₂ concentrations are high (polluted regions). The dissolution of SO₂ in water, most particularly sea water (which is alkaline), represents a further important sink.

Acidification

The production of sulfuric acid in air masses over industrialized continents led to the acid rain problem. This acidification was recognized by the mid-nineteenth century, initially as emissions of hydrochloric acid from the alkali and soap industry, but later coal combustion was also seen as an important source of acid. By the 1950s, long-term records of rainfall chemistry showed increasing fluxes of sulfuric acid to the ground. In Scandinavia, poorly buffered soils began to respond to the effects of acid rain and the pH of lakes declined, resulting in a loss of fish. Through the 1970s and more particularly the 1980s, acid rain became a keenly fought environmental issue, which led to a range of protocols in North America and Europe that served to limit the release of sulfur into the atmosphere.

Acids accumulated over the winter are concentrated into the first meltwaters of spring and send an acid shock through the environment. Young fish are readily injured, but amphibians such as frogs and salamanders also suffer. More generally the delicate ecological web that depends on the freshwater systems is affected. The debate over acid rain also examined the effects on forests. Forest damage arises from a complex array of factors that include air pollution (ozone, SO_2 , acid rain, etc.) along with climate stress (i.e., drought), forest management practices, the age of stands, etc. Acidified rain could change the nutrient balance of soils (magnesium, potassium, and calcium) and mobilize toxic metals (aluminum, cadmium, zinc, etc.).

Damage to buildings is frequently seen as an important impact of acid rain, perhaps because we confront our urban fabric almost every day. It is exposed to the ravages of high concentrations of air pollution, which means that buildings have long been disfigured by sooty crusts. Hidden underneath these crusts damage takes place through the oxidation of deposited SO_2 . Stones build up thick gypsum (calcium sulfate) layers and metals are also prone to attack. In general, the deterioration of metal work and, to a lesser extent, stone has declined as SO_2 concentrations have decreased in urban air.

There is now a decrease in the sulfur deposited in rain over Europe. This has led to complaints that some agricultural crops are now sulfur deficient. Although acid rain provided sulfur for crops, it is obviously important to remember its large-scale harmful effects before regretting its gradual elimination. Even though sulfur deposits have decreased in Europe, the acidity of rain has not always fallen to match this decline. Increases in nitric acid in air (from nitrogen oxides produced during combustion) and a decline in the alkaline content of the atmosphere may account for

this. Today the acidification of rain has shifted geographically to tropical regions with fast-growing economies. Here there are new problems on ecosystems and land types quite different from those affected in Europe and North America.

Aerosol Sulfur and Climate

Sulfate in the atmosphere is present some of the time in acid aerosols, most commonly sulfuric acid or ammonium bisulfate. The sulfuric acid aerosol is typically present at diameters of 2–3 μm . This makes the particles excellent cloud condensation nuclei (CCN), but also a potential threat to human health when they are found concentrated in urban air. Sulfates, nitrates, and methanesulfonates are frequently found at enhanced concentrations on sea salt aerosols as these strong acids can displace as much as 90% of the chloride as hydrogen chloride, which degases to the atmosphere.

Sulfate aerosols appear to have an important role in controlling climate. The CO_2 -driven greenhouse effect provides a link between increases in global temperature and rising emissions of CO_2 from fossil fuels. SO_2 is also a product of much fossil fuel use and results in the presence of sulfate particles in the atmosphere. These particles are likely to increase the number of CCN and hence the reflectivity of clouds. This would enhance the albedo of the Earth and lessen the temperature increase imposed by anthropogenic CO_2 emissions. The cooling effect, although difficult to estimate, is only a fraction of the increase from greenhouse gases.

Aerosols are incorporated into the well-known Gaia hypothesis, which sees the Earth as a self-regulator entity, particularly in terms of its temperature. One of the main ways in which atmospheric processes have been interpreted within the Gaia concept has been the potential for DMS emissions to stabilize temperature, against changes imposed by shifts in solar irradiance (Figure 3). This is known as the CLAW hypothesis, after Charlson, Lovelock, Andreae, and Warren, who first suggested it in 1987. It sees long-term climate as part of a feedback loop in which warmer temperatures would increase phytoplanktonic activity in the oceans. This in turn would increase emissions of DMS. Higher DMS concentrations in the atmosphere would increase sulfate containing CCN and thus cloudiness and albedo. This would lead to lower temperatures. It is an attractive and popular vision of the biological mediation in global climate, although proof from ice cores has yet to offer convincing proof for this concept of climate control.

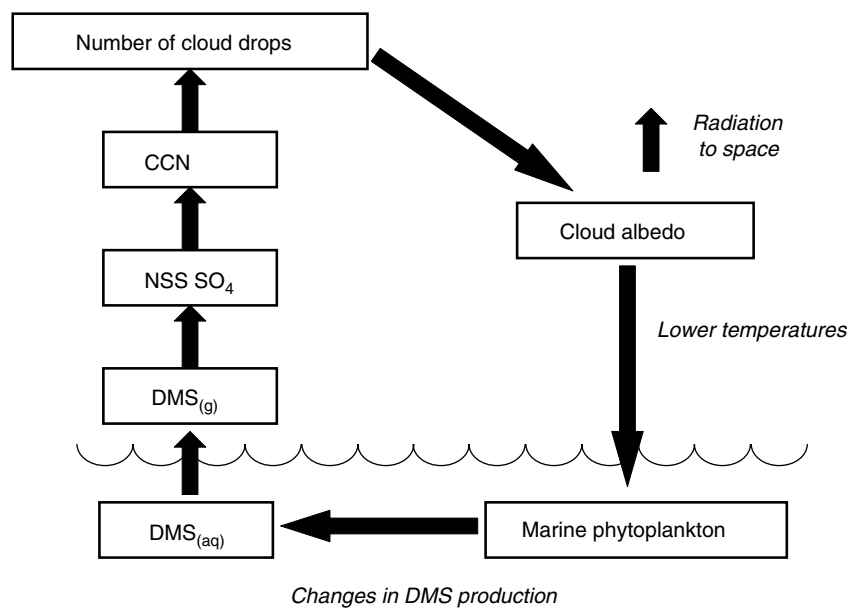


Figure 3 Schematic representation of the CLAW hypothesis. Here phytoplanktonic activity in the oceans affects the emissions of DMS. Its oxidation can form sulfate-containing CCN and increasing cloudiness and albedo, which would lead to lower temperatures. If these lower temperatures cause the phytoplankton to produce less DMS, climate could effectively be controlled.

Despite the size of the natural sulfur reservoirs, combustion of fossil fuels, mining, agriculture, and the drainage of large saline basins such as the Aral Sea have had a major impact on the sulfur cycle. Wind-blown sulfate dusts and acid rain have increased, and although emissions may have shifted away from the North Atlantic sector in recent decades, anthropogenic impacts on sulfur cycles will remain an important consideration in developing regions for some time to come.

Further Reading

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